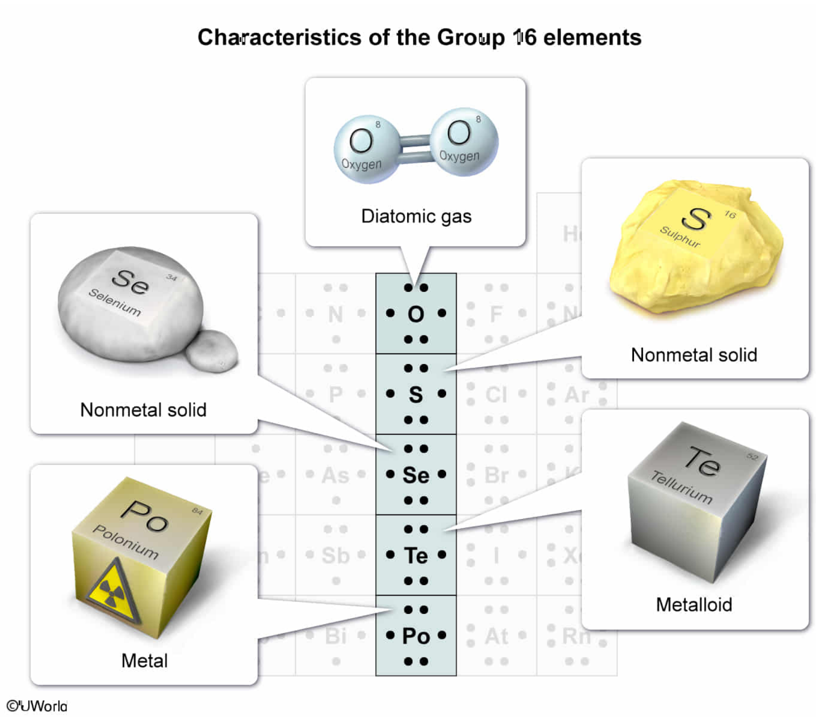
In elements 1–86 on the periodic table (at 25 °C), one particular group contains one element that forms a diatomic gas in its elemental state, at least one metalloid, two nonmetal solids that are not metalloids, and one element that is exclusively radioactive. The group is:

- A. Group 14. [6%]
- B. Group 15. [27%]
- C. Group 16. [54%]
- D. Group 17. [10%]

Correct
55% answered correctly

Explanation:



Elements are **arranged in groups** on the periodic table based on their electronic configurations. Elements within the same group have the same number of valence electrons and share similarities in their **chemical properties** (ie, reactivity, bonding) as a result. However, elements within a group can have very different **physical properties**.

In Group 16 (ie, the oxygen group, also called the chalcogens), each element has six valence electrons and tends to gain two additional electrons during chemical bonding. However, the physical properties of each of these elements are very distinctive.

Oxygen atoms form double covalent bonds with other oxygen atoms to achieve a full valence shell of electrons; therefore, oxygen is one of the seven common elements that exist as a diatomic molecule (O_2) in their elemental state.

Sulfur (S) is a dull yellow, nonmetal solid, and selenium (Se) is a lustrous gray, nonmetal solid. Both elements have very low electrical conductivity and lack the necessary extent of semiconducting ability required to be considered a metalloid. In contrast, tellurium (Te) is a semiconductor and meets the necessary characteristics to be classified as a metalloid.

Polonium (Po) is very rare, but all isotopes of polonium are unstable, causing polonium to exist exclusively as a radioactive element.

Therefore, in elements 1–86 on the periodic table (at 25 °C), Group 16 has one elemental diatomic gas, one metalloid, two nonmetal solids that are not metalloids, and one element that is exclusively radioactive.

(Choice A) Group 14 contains only *one* nonmetal solid that is not a metalloid (C), *two* metalloids (Si and Ge), and two metals (Sn and Pb).

(Choice B) Group 15 has one elemental diatomic gas (N_2) but has *one* nonmetal solid that is not a metalloid (P), *two* metalloids (As and Sb), and one metal (Bi).

(Choice D) At 25 °C, F_2 and Cl_2 are gases, Br_2 is a volatile liquid, and I_2 is a nonconducting solid. As such, Group 17 contains *two* elemental diatomic gases (F_2 and Cl_2), *one* nonmetal solid that is not a metalloid (I), and *one* metalloid that is also found exclusively as a radioactive element (At).

Educational objective:
Elements are arranged into groups on the periodic table based on their electronic configurations. Elements within the same group have the same number of valence electrons (similar chemical properties) but can have very different physical properties.

Time spent: 0 sec

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Subject:
General Chemistry

System:
Atomic Theory and Chemical Composition